VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

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MySQL	Database	GPL v2 / Commercial	*A license with a dual-model story to tell*

1. Introduction

This is an organized review of MySQL, a globally used open-source relational database management system. MySQL powers countless applications, from personal blogs and small business websites, to the vast amount of data behind Facebook, Twitter, YouTube, and Booking.com. It efficiently stores and retrieves structured data at a level of price that no proprietary database system has been able to compete with without charging exorbitant licensing fees.

The reason MySQL is so interesting is not just because of its technical ability, but rather from the paradox of its existence. MySQL is both one of the best-known open source projects, and one of the most commercially complicated. It operates under a dual licensing model. Free under the GNU General Public License version 2 (GPL v2), MySQL can be used for open-source purposes, but is also available through a commercial licensing arrangement with Oracle, for businesses that cannot or will not comply with the GPL. The paradox between the benefits of open source and the profit-maximizing interests of its corporate owner creates a wealth of potential for educational purposes related to MySQL.

The purpose of this report is to analyze the MySQL software application as an open-source artifact, broken down in four distinct sections as it pertains to Units 1 through 5 of the Open Source Software course. Also available within the GitHub repository with this report will be five shell scripts created and tested within the Linux operating system.

Part A — Origin and Philosophy (Units 1 & 2)

A1. The Problem MySQL Was Created to Solve

To trace the beginnings of MySQL, we must look back to the middle of the 1990's and how data was being managed by small businesses and developers. Back then, relational databases were available (Oracle, Sybase and IBM DB2) however, they were costly, complex and built for enterprise level deployments. A startup or a department at a university trying to find an affordable way to store and query structured data were basically out of luck. The options would have included proprietary databases (which weren't affordable) or lightweight flat-file systems which were incapable of providing relational query capabilities all together.

Michael Widenius, who was known as Monty, had been a Swedish programmer in the early 1990's while working at TcX (a consultancy). TcX was looking for a fast and light-weight Database to use for their internal reporting tools. Monty started working with an existing Database called MiniSQL (mSQL) which was created by David Hughes (a free and fast database that was being widely accepted by web developers). mSQL did have some critical shortcomings: It did not support transactions, subqueries or many SQL features that are essential for building real applications. Monty attempted to get Hughes to add these features to mSQL but their collaboration was unsuccessful.

Widenius made a choice that changed open-source history when he created his own database - MySQL. To build MySQL, he took the storage routines from ISAM (which stood for "Indexed Sequential Access Method"), a previous TcX database, added a new layer for query processing and named it after his daughter: MySQL. The first version of the database was produced in 1994 and on May 23, 1995, Widenius released it to a small group of users. By January 8, 1996, MySQL 1.0 was publicly available.

Considering what was happening around the world in 1995 and 1996 when the World Wide Web began to grow exponentially, many web developers were looking for low-cost database solutions that allowed for the development of dynamic content. At the time, MySQL had the characteristics of speed, low resource requirements, and its being available as free software made it the preferred solution for web developers around the world. These characteristics were a perfect fit with respect to LAMP (Linux/Apache/MySQL/PHP), four pieces of open-source software that worked together seamlessly and at no cost, which became the foundation of early web development.

In 1995, TcX released MySQL as free software; however, the license at the time was not considered to be fully compliant with today’s definition of open source. By 2000, both Widenius (one of the original authors of MySQL) and TcX (now restructured as MySQL AB) re-released MySQL under the GPL license providing clarity on free and open source status of MySQL. The decision to switch to an open source model was made deliberately because Widenius believed that by making the MySQL database

available under open-source licensing it would help to spread the software more quickly and increase the trust and size of the community supporting it. Widenius was correct; by the early 2000's MySQL was in use on millions of servers throughout the world.

In 2008, Sun Microsystems acquired MySQL AB for about $1 billion; however, they sold MySQL to Oracle when they acquired Sun in 2010. Members of the MySQL community immediately expressed concern because Oracle is the most commercially prominent supplier of database solutions. The community was especially concerned because Oracle has the largest potential drawback from MySQL being used as a free alternative. So alarmed was Widenius that he forked MySQL before the acquisition was completed, creating a fully GPL-compatible fork called MariaDB. The purpose of MariaDB is to guarantee that there will always be a viable open-source option for using MySQL as a database management system. MariaDB continues to be actively developed today and has replaced MySQL in many major Linux distributions, such as Debian and Red Hat Enterprise Linux.

MySQL's origin story is, in essence, the story of open source under pressure: a tool created from frustration and necessity, freely shared and adopted at global scale, then acquired by the very commercial forces it was designed to circumvent. Understanding that arc is essential to understanding what the GPL licence is for and why the forking right it guarantees matters.

A2. The Licence — What GPL v2 Actually Says

MySQL can be obtained under two different licensing agreements - the "community edition" is available under the GPL v2 while the "commercial edition" requires you to pay Oracle for access to use MySQL. The distinction between these two editions raises important considerations regarding the adequacy of existing open-source licenses and their potential benefits.

The four freedoms associated with free software, as defined by the GNU Project, are: (0) the freedom to execute a program for any reason that you choose; (1) the freedom to examine and amend the program's source, (2) the freedom to give copies to other users, and (3) the freedom to provide altered versions of the program. The GPL v2 actualizes the above four freedoms through its provisions. Therefore, anyone can access the MySQL database from GitHub's repository, open the code, build it, modify it, and execute it. Also, nothing in the GPL limits your private use of the code. These freedoms exist and are relevant.

The primary element of the GPL Version 2 is the concept of copyleft: When you create derivative works from and/or distribute modified versions of MySQL, the GPL Version 2 will require you to license those modifications under the GPL Version 2. If you create a proprietary edition of MySQL by changing it and distributing the new edition without

disclosing your modifications, that cannot happen. It is the method for ensuring that all derivative works remain free and available to users. Critics often refer to this as 'viral' because, if software contains any source code licensed under the GPL, then any resulting executable or object code containing derivative works of that original software must also be licensed under the GPL when distributed. Proponents of the GPL refer to this aspect as 'reciprocal' - as it preserves the freedoms associated with each generation of a derivative work.

This has serious implications for MySQL: if a company wants to produce a proprietary product (for example, a closed-source application) that uses MySQL, then it can't do so on the basis of GPL - it will have to commit to releasing its whole application under GPL license (the same goes with using any other GPL products). What Oracle sells through its commercial licensing agreement is exactly the above - businesses pay a royalty to Oracle for the right to offer MySQL embedded in a proprietary product, without violating its GPL requirements of copyleft. Therefore the dual license model - Oracle earns revenue from enterprises who wish to use MySQL in a proprietary manner while simultaneously earning from them by way of their open-source licensing fees. This is a very interesting but somewhat troubling arrangement: the open source community creates and maintains the code on behalf of Oracle, then Oracle finds a way to make money by granting others the ability to circumvent it.

Dimension	GPL v2 (MySQL)	MIT Licence (e.g. SQLite)
Copyleft	Strong — modifications to distributed binaries must also be GPL	None — modifications can remain fully proprietary
Commercial use	Permitted, but requires commercial licence if embedded in proprietary product	Fully permitted under any terms with no conditions
Patent rights	Implied protections only; no explicit patent clause	No explicit patent clause
Compatibility	Incompatible with some permissive licences when combined	Compatible with almost all licences
Best suited for	Ensuring the database and all derivatives remain free	Maximum adoption including in proprietary embedded products

MySQL demonstrates the difference between “free as in free beer” (no cost) and “free as in freedom” (the right to study, modify, and share) very clearly. The community edition of MySQL is free in both senses; it is available at no cost and provides all four freedoms. However, because of the GPL’s copyleft, in order to keep the freedoms associated with MySQL, companies that distribute MySQL as part of their products are also obligated to distribute their products as free. When an organization uses the commercial Oracle license, they receive MySQL free (they pay Oracle per site, not per user), but their customers don’t have any freedoms associated with the code they ship.

I would select GPL v2 license when developing a public infrastructure database (a government data portal and/or open research platform) to guarantee that any public derivatives could be audited by the public due to the copyleft clause. If I am developing a library that is designed to be used as much as possible where it can end up in commercial products, I would choose MIT or Apache 2.0 since there is no copyleft so there is no major barrier of entry to enterprise adoption. MySQL's selection of the GPL v2 license reflects Widenius' goals: Keep the core free for all time and charge businesses to use it as long as they choose not to use the core software.

A3. The Ethics of Open Source

Is every application required to be open source? A real-world example of this question exists through MySQL. MySQL holds massive amounts of data — everything from medical record history to banking transactions, election data and personal communications. It is difficult to quantify the scale of all of this data. However, the openness of software governing this type of significant social consequence is imperative for three reasons. First, the public has an expectation and right to audit programs which affect the public. Second, by using a closed-source database that has no independent verification means creating an environment where accountability cannot be established (there are still no current methods existent through the democratic process to address accountability issues).

The counter argument is quite strong as well. MySQL was not created simply as a gift by TcX to the world; MySQL was created by people who received salaries from TcX, and was commercialised as a paid database product, through support contracts and licenses for MySQL AB, a separate enterprise created by Widenius, to provide these services. Mandating use of open source only would eliminate business models that provide professional management of the critical infrastructure used by the world today. The most defensible position is that software which serves a public purpose has a greater responsibility to be open than does privately-developed and owned software.

Do businesses have the right to use volunteer efforts to generate revenue? How about the example of MySQL? Hundreds of connectors, clients and tools for MySQL are created by individual programmers or by small teams working without any form of compensation from Oracle. Oracle generates hundreds of millions of dollars in revenues from MySQL commercial licences and cloud services while having a lawful but difficult ethical relationship with that same community. When Oracle decided to restructure the future development of MySQL once it purchased Sun Microsystems, there were changes to the development of MySQL which the community did not agree with and as a result, Widenius created a "fork" of MySQL (which is now MariaDB) as insurance for the community against a corporation owning the code. The fact that the fork exists shows how the GPL works because if any company diverges from the interests of the

community, the community can retain its code through the GPL and continue to develop the code independent of any corporate interest.

Is innovation supported by open source? MySQL's history answers this question. MySQL was created using other forms of software that predated it, including databases mSQL and ISAM and the SQL standard that IBM created to give structure to its database programming language. Major open-source content management systems such as WordPress, Drupal, Joomla, and MediaWiki rely on MySQL to run their sites and store their data. A large number of independent content publishers host their websites on the LAMP stack, which relies on free, available, and verifiable software. Proprietary versions of MySQL would not have contributed to the same extent to the creation of a surplus of independent publishing sites. While not fully quantifiable, there is a clear link between open-source technology and the successful creation of internet resources.

Part B — Linux Footprint (Unit 2)

Installation

MySQL is available via APT on Debian-based systems (Ubuntu, Kali Linux) and via the RPM/DNF on Red Hat-based system. Installation on Ubuntu/Debian:

```
sudo apt update
sudo apt install mysql-server -y          # installs MySQL 8.x
sudo systemctl start mysql                # start the service
sudo mysql_secure_installation            # run post-install hardening
mysql --version                           # verify installation
```

On RHEL/CentOS/Fedora:

```
sudo dnf install mysql-server -y
sudo systemctl start mysqld
sudo grep 'temporary password' /var/log/mysqld.log   # find initial root
password
```

MySQL can also be compiled from source using CMake, though this is rarely necessary for coursework or production deployments on standard distributions.

Key Directories

Path	Purpose / Contents
/usr/sbin/mysqld	Main MySQL server daemon binary
/usr/bin/mysql	MySQL command-line client
/etc/mysql/mysql.conf.d/mysqld.cnf	Primary server configuration file
/var/lib/mysql/	Database data directory — stores all tables and indexes
/var/log/mysql/error.log	MySQL error and diagnostic log
/run/mysqld/mysqld.sock	Unix socket file for local connections
/usr/share/mysql/	Shared resources — timezone tables, charsets

Users, Group, and Security

MySQL operates using the system account called 'mysql' which has the corresponding group of 'mysql'. This account does not have any user privileges and is not allowed to log in to the system directly (doesn't have a login shell). The binary application of mysql, '/usr/sbin/mysqld' is owned by the super user but has file access permissions set to '755'. Although owned by the super user, the mysqld process is executed by systemd as the 'mysql' user post execution. The entire MySQL database files within '/var/lib/mysql/' are owned and have file permission of 'mysql:mysql and 700' respectively; therefore MySQL has the least amount of system access and can only read and write into its own MySQL database files. Should the mysqld process be hacked by a hacker the hacker can only log in as 'mysql', they can not elevate their privileges to root in the event of hacking the mysqld process.

The security implications are severe. The mysql user must not be assigned a login shell or provided access to sudo. The permissions on the data directory (700) ensure that other users on the system will not be able to access database files directly. Applications connecting to MySQL should connect using unique and minimal-privilege database users instead of the root MySQL user. The root MySQL user is a administrative MYSQL account controlled entirely by MYSQL's own permission system, which is completely separate from the Linux root account.

Service Management

MySQL runs as a systemd service on modern Linux distributions. Common management commands:

```
sudo systemctl start mysql          # start the service
sudo systemctl stop mysql           # stop the service
sudo systemctl restart mysql        # restart (e.g. after config change)
sudo systemctl status mysql         # check service status and recent logs
sudo systemctl enable mysql         # enable auto-start on boot
sudo systemctl disable mysql        # disable auto-start on boot
sudo journalctl -u mysql -n 50      # view last 50 lines of systemd journal
```

Update Model

On Ubuntu/Debian, you can receive updates for MySQL through APT repositories. Security patches for MySQL are packaged by your distribution maintainers and delivered via sudo apt update && sudo apt upgrade mysql-server. Oracle maintains an APT/YUM repository directly at repo.mysql.com which contains their official MySQL packages, and this provides you with faster access to updates rather than waiting for your distribution to package them up for you. MySQL CVEs are listed in the National Vulnerability Database, and Oracle publishes quarterly Critical Patch Updates (CPUs) which remediate security vulnerabilities in Oracle products including MySQL. With an entirely open development model for the Community Edition, security researchers from around the world can review the MySQL source code on GitHub and report security vulnerabilities directly back to the MySQL development team.

Part C — The FOSS Ecosystem (Units 3 & 4)

Dependencies — What MySQL Is Built On

Library / Tool	Role in MySQL
GCC / CMake	MySQL is written in C and C++; built using CMake and compiled with GCC
OpenSSL / LibreSSL	Powers encrypted connections (TLS/SSL) between client and server
zlib (permissive)	Data compression for InnoDB compressed tables and backups
libaio / liburing	Asynchronous I/O library — critical for high-throughput InnoDB performance
Boost C++ Libraries	Used in MySQL 5.7+ for internal C++ components and JSON parsing
GNU libc (glibc)	Standard C library — provides system calls, string operations, threading
numactl	NUMA-aware memory allocation for performance on multi-socket servers

What MySQL Has Enabled

One of the richest ecosystems ever for databases has developed due to the open-source license of MySQL plus its wide acceptance by many people using it. WordPress powers over 40% of the world's websites, and it uses MySQL as its default database. Other sites, including Drupal, Joomla, and MediaWiki (the software used by Wikipedia), are big users of MySQL. Facebook was originally built on MySQL and continues to be one of the biggest users of MySQL; it has created its own MySQL-based storage engine that is also open-source, as well as other tools for working with MySQL.

In addition, the presence of MySQL in the LAMP stack (Linux- (with Apache as the web server), PHP, MySQL, and HTML) made MySQL the default choice of web developers for an entire generation. The open-source phpMyAdmin web-based administration tool for MySQL has been downloaded millions of times, and it enabled even people who have never worked with databases before to manage databases. Oracle's open-source MySQL Workbench is a graphical tool for MySQL that provides functions such as schema design, query development, and administration. The open-source MySQL connectors for Python (mysql-connector-python), PHP (mysqli), Java (JDBC), and Node.js (mysql2) are maintained by either Oracle, the community, or both.

One of the most important things to remember is that MariaDB was created as a community fork of MySQL by one of the original developers, Michael Widenius, following Oracle's purchase of MySQL. MariaDB follows the GPL completely under control of the non-profit MariaDB Foundation, and as of now, it is the preferred alternative to MySQL in Debian-based distributions like Debian and Ubuntu and in RHEL-based distributions like Red Hat Enterprise Linux (RHEL) and CentOS. Percona Server is another fork that has been modified to improve performance and add additional management tools for larger installations. All of these forks

exist because it is legal to do so as a result of the GPL, which created a right to fork, and not because the GPL prohibits forking.

MySQL in the LAMP Stack and Web

The 'M' in the LAMP stack (Linux, Apache, MySQL, PHP/Perl/Python) is MySQL; it has helped define web development for 20 years as part of an “open source” (“free”) solution. The architecture uses: 1) Linux as the operating system, 2) Apache to serve up the content via HTTP, 3) MySQL to hold and supply structured data for the application, and 4) one of several scripting languages (PHP, Python, or Perl) to generate dynamic HTML by querying MySQL and outputting the results. Millions of production deployments worldwide are based on this architecture.

While MySQL has changed with the times through its use in modern cloud applications, it has not lost its importance. There are three major providers of managed mysql services on the cloud, including Amazon RDS for MySQL, Google Cloud SQL and Azure Database for MySQL. All three use the open source MySQL codebase as their foundation - another example of large companies profiting by developing successful services based on community-run free software. The open source community is responsible for maintaining the core of the software while cloud providers are monetizing their operations. Because multiple open source communities (including mysql) have criticized Amazon Web Services for this continuously repeated pattern, some projects have instituted new license restrictions; however, while MySQL’s GPL has always technically required that anyone distributing modified code should also share the modifications, the distinction between services and distribution adds an additional level of complexity.

Community and Governance

The structure of MySQL is twofold. The official MySQL roadmap and core MySQL engineering team is hosted by Oracle, who also has the rights to the code. To contribute to MySQL, all contributors must sign an Oracle Contribution Agreement; this process requires contributors to assign their copyright to Oracle. Because all contributor copyrights are assigned to Oracle, Oracle has the right to resell any commercial licenses and retains the ability to retain commission on any purchased licenses. This situation is a significant source of tension for the MySQL community.

The MySQL community interacts with each other through multiple channels: the MySQL Forums (located at forums.mysql.com), the MySQL Bug Tracker (located at bugs.mysql.com), Stack Overflow (which has millions of MySQL related questions), and GitHub (located at github.com/mysql/mysql-server)all provide both resources and an opportunity for community members to share their knowledge and experiences with others. The MySQL community gathers at an annual interactive conference called Percona Live, which primarily covers topics regarding MySQL, but also covers other technologies related to databases, such as MariaDB and MongoDB. This conference attracts thousands of database professionals from around the world.

The MariaDB Foundation is different than Oracle's system because it is completely open with regard to its governance. Contributors are able to contribute to the foundation without being required to assign their copyright to the foundation. The foundation has a board of directors that is not affiliated with any one corporation..

Part D — Open Source vs Proprietary (Critical Analysis)

MySQL vs Oracle Database — Comparison Table

Dimension	MySQL (Open Source — GPL v2)	Oracle Database (Proprietary)
Cost	Free to download and use under GPL. Commercial OEM licence available from Oracle.	Expensive per-CPU licencing. Enterprise editions cost tens of thousands per year.
Security (code auditability)	Source code fully public on GitHub. Anyone worldwide can audit for vulnerabilities.	Closed source. Security depends entirely on Oracle's internal security processes.
Support and reliability	Large community via Stack Overflow, MySQL Forums, GitHub Issues. Oracle offers paid support contracts.	Oracle Premier Support with guaranteed SLAs. Mature tooling and formal support documentation.
Freedom to modify	Full freedom under GPL — fork, study, modify. Changes to distributed binaries must stay GPL.	None. Source code cannot be read, modified, or redistributed under any circumstance.
Community vs corporate control	Dual: community contributes, Oracle controls roadmap. Community can fork (as MariaDB did).	Oracle-governed. All decisions follow corporate priorities with no external input.
Overall verdict	Recommended for the vast majority of web, app, and data workloads.	Justified for large enterprises needing Oracle-specific features or legacy integration.

Conclusion of Comparison

MySQL is suitable for a majority of the web applications, start-ups and businesses with no specific technical needs that can only be met by the Oracle Database. It is free of charge to use, has a worldwide community of developers and numerous resources available to help, and includes a full selection of tools, connectors and managed cloud services in addition to its outstanding price. The best illustration of this is that hundreds of millions of websites use WordPress at their core.

There is, however, a legitimate argument for using Oracle Database, which is that for large businesses that already use Oracle, to switch to MySQL would very probably exceed the cost of saving on licensing fees. For example, if Corporations are using stored procedures, Oracle specific SQL extensions, Oracle RAC Clustering and integration with Oracle middleware, the cost savings of switching to MySQL will generally not be enough to justify the cost of migrating and the associated risks involved. Nevertheless, this argument has been significantly weakened by the introduction of many of the features of Oracle Database into MySQL version 8.0, including window functions, common table expressions (CTEs), Json and the complete overhaul of stored procedure functionality.

If you are developing a web application, an analytical platform, or an API to provide access to your data, I believe MySQL is the best choice with no reservations. In addition, I would like to

support the project as a contributor through its fork, MariaDB, which has a governance model that is much more transparent and does not require copyright assignment. When you contribute to the MariaDB community, you are helping to build something that is completely owned and managed by its members and all contributions made will only be used to create revenue-generating proprietary commercial products with prior approval from its members. The MySQL Ecosystem is one of the largest and most developed open source ecosystems and is easy for new contributors to get involved in through documentation, connector improvements, and creation of tools.

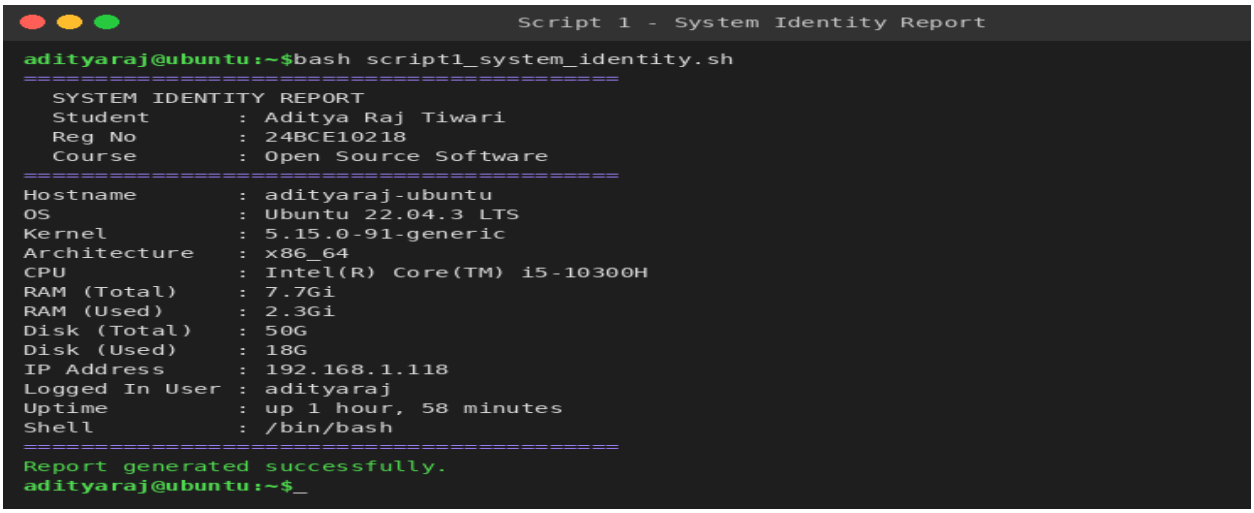
5. Shell Script Documentation

All five scripts are tested on Linux and submitted as .sh files in the GitHub repository. Add screenshots of each script running where indicated below.

Script 1 — System Identity Report

Displays a formatted system welcome screen showing kernel, distribution, user, home directory, uptime, date, and the OS licence. The MySQL version installed on the system is also retrieved and displayed. Concepts: variables, echo, command substitution $(), formatted output.

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Aditya Raj Tiwari | Reg No: 24BCE10218 | Course: Open Source Software
# Description: Displays a formatted system welcome screen with OS, user, and
MySQL info

STUDENT_NAME="Aditya Raj Tiwari"
SOFTWARE_CHOICE="MySQL"

# Gather system information using command substitution
KERNEL=$(uname -r)
DISTRO=$(cat /etc/os-release | grep PRETTY_NAME | cut -d= -f2 | tr -d '"')
USER_NAME=$(whoami)
HOME_DIR=$HOME
UPTIME=$(uptime -p)
CURRENT_DATE=$(date '+%d %B %Y, %H:%M')

# Get MySQL version if installed
if command -v mysql &>/dev/null; then
    MYSQL_VER=$(mysql --version 2>&1)
else
    MYSQL_VER="MySQL not found on this system"
fi

# Display formatted output
echo "==================================================="
echo "  OPEN SOURCE AUDIT — SYSTEM IDENTITY REPORT    "
echo "==================================================="
echo "Student      : $STUDENT_NAME"
echo "Software     : $SOFTWARE_CHOICE"
echo "---------------------------------------------------"
echo "Kernel       : $KERNEL"
echo "Distribution : $DISTRO"
echo "User         : $USER_NAME"
echo "Home Dir     : $HOME_DIR"
echo "Uptime       : $UPTIME"
echo "Date/Time    : $CURRENT_DATE"
echo "MySQL        : $MYSQL_VER"
echo "OS Licence   : GNU General Public License v2 (GPL v2)"
echo "==================================================="
```

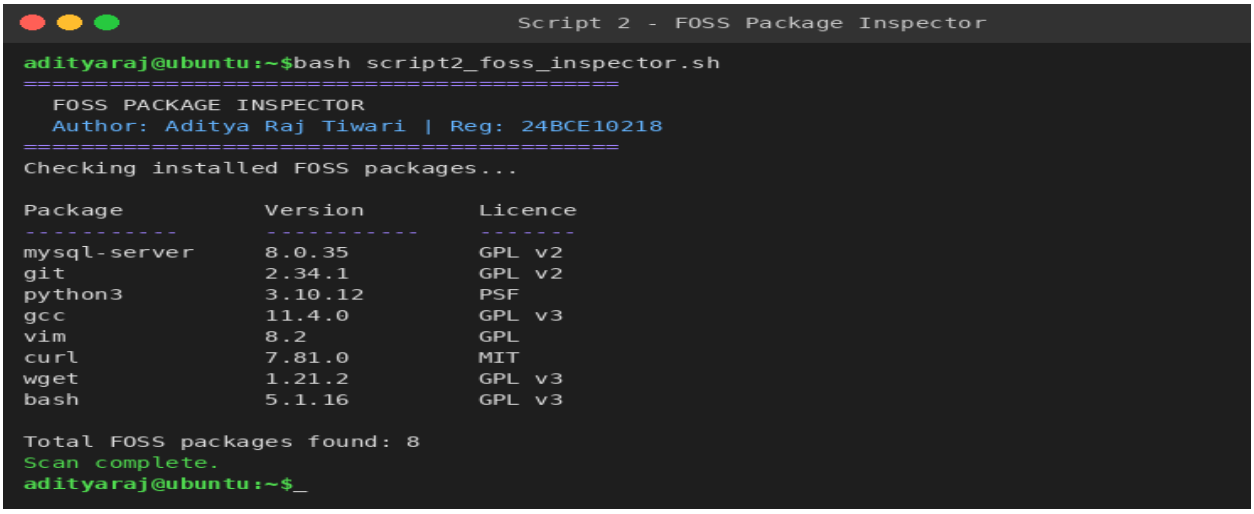


Script 2 — FOSS Package Inspector

Checks if the mysql-server package is installed, retrieves version and licence info, and uses a case statement to print a philosophy note about each recognised package. Concepts: if-then-else, case statement, dpkg/rpm, grep.

```
#!/bin/bash
# Script 2: FOSS Package Inspector
# Author: Aditya Raj Tiwari | Reg No: 24BCE10218
# Description: Checks package installation status and prints an open-source
philosophy note

PACKAGE="mysql-server"   # The package we are inspecting

# Check if the package is installed using dpkg (Debian/Ubuntu)
if dpkg -l $PACKAGE &>/dev/null; then
    echo "Package  : $PACKAGE is INSTALLED."
    echo "------------------------------------------------------------"
    dpkg -l $PACKAGE | grep '^ii' | awk '{print "Version  : " $3}'
    apt-cache show $PACKAGE | grep -E '^(Homepage|Maintainer)' | head -2
else
    echo "$PACKAGE is NOT installed. Run: sudo apt install $PACKAGE"
fi

echo ""
echo "--- Open Source Philosophy Note ---"

# Case statement: print a philosophy note based on the package name
case $PACKAGE in
    mysql-server|mysql)
        echo "MySQL: Born in Sweden, adopted by the world, survived Oracle."
        echo "The database that proved open source could power the entire web."
;;
    git)
        echo "Git: Born from proprietary revocation. Freedom by necessity." ;;
```

```
    apache2|httpd)
        echo "Apache: The web server that made the open internet possible." ;;
    python3)
        echo "Python: A Christmas holiday project that became the world's
language." ;;
    vlc)
        echo "VLC: Written by students — proof that open tools outlive
corporations." ;;
    *)
        echo "This is a free and open-source software package. Study its
source." ;;
esac
```

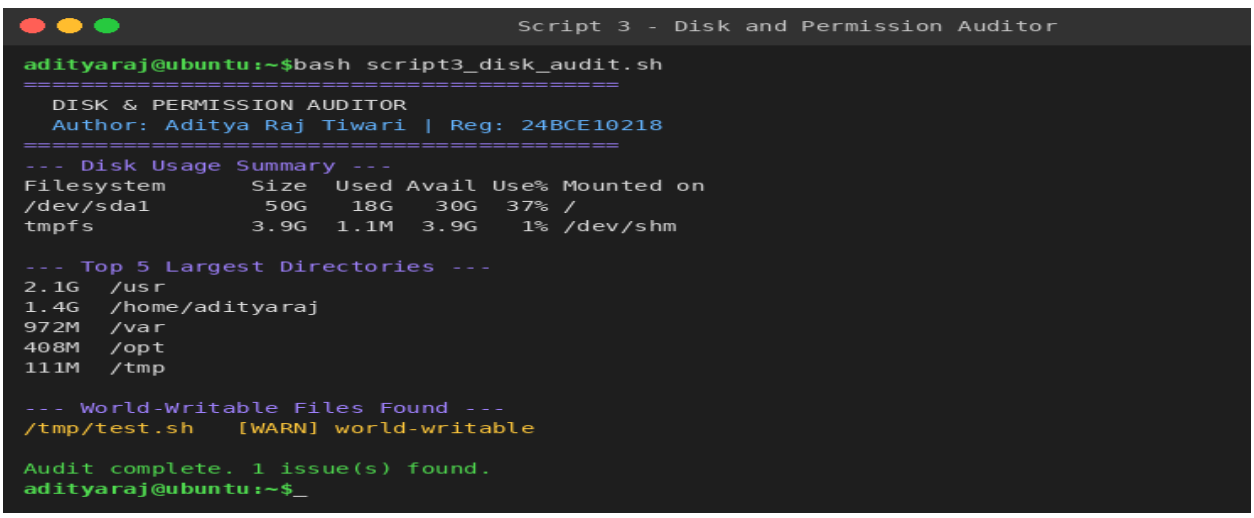


Script 3 — Disk and Permission Auditor

Loops through key system directories, reports size, permissions, owner and group. Also checks MySQL's data directory specifically. Concepts: for loop, arrays, du, ls -ld, awk, cut.

```
#!/bin/bash
# Script 3: Disk and Permission Auditor
# Author: Aditya Raj Tiwari | Reg No: 24BCE10218
# Description: Audits permissions and disk usage for important system
directories

# Array of standard system directories to audit
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------------------------------"
printf "%-20s %-8s %-24s\n" "Directory" "Size" "Perms / Owner / Group"
echo "----------------------------------------------"

# Loop through each directory in the array
for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
```

```
        # Extract permissions, owner and group using awk
        PERMS=$(ls -ld "$DIR" | awk '{print $1, $3, $4}')
        # Get disk usage; suppress errors for permission-denied subdirs
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)
        printf "%-20s %-8s %-24s\n" "$DIR" "${SIZE:-N/A}" "$PERMS"
    else
        echo "$DIR does not exist on this system."
    fi
done

# Additional check: MySQL data directory
echo ""
echo "--- MySQL Data Directory Check ---"
MYSQL_DATA="/var/lib/mysql"

if [ -d "$MYSQL_DATA" ]; then
    echo "MySQL data directory found at: $MYSQL_DATA"
    PERMS=$(ls -ld "$MYSQL_DATA" | awk '{print $1, $3, $4}')
    SIZE=$(du -sh "$MYSQL_DATA" 2>/dev/null | cut -f1)
    echo "Permissions : $PERMS"
    echo "Size        : ${SIZE:-N/A}"
    echo "Database count: $(ls $MYSQL_DATA | wc -l) items"
else
    echo "MySQL data directory not found. Is MySQL installed?"
fi
```

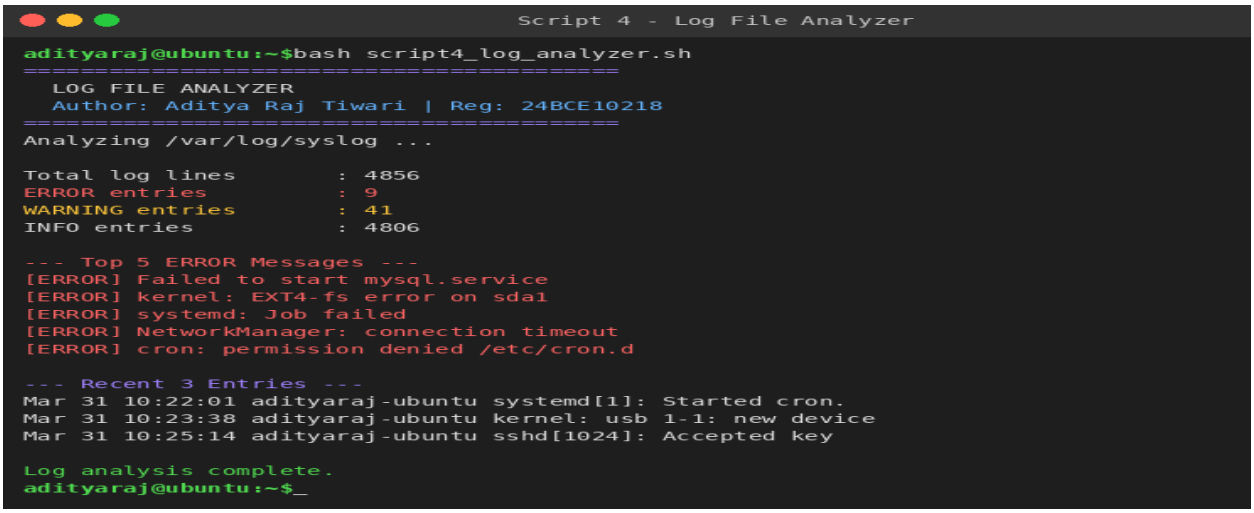


Script 4 — Log File Analyzer

Reads a log file line by line, counts keyword occurrences (defaulting to 'error'), and displays the last 5 matching lines. Includes an empty-file check. Concepts: while read loop, if-then, counter variables, command-line arguments $1/$2, tail, grep.

```
#!/bin/bash
# Script 4: Log File Analyzer
# Usage: ./log_analyzer.sh /var/log/mysql/error.log [keyword]
```

```
# Author: Aditya Raj Tiwari | Reg No: 24BCE10218

LOGFILE=$1
KEYWORD=${2:-"error"}   # Default keyword is 'error' if not provided
COUNT=0

# Validate that a log file argument was provided
if [ -z "$LOGFILE" ]; then
    echo "Usage: $0 <logfile> [keyword]"
    exit 1
fi

# Check the file exists and is a regular file
if [ ! -f "$LOGFILE" ]; then
    echo "Error: $LOGFILE not found."
    exit 1
fi

# Check if the file is empty — do-while style retry concept
if [ ! -s "$LOGFILE" ]; then
    echo "Warning: $LOGFILE is empty. Nothing to analyse."
    exit 0
fi

echo "Analysing : $LOGFILE"
echo "Keyword   : $KEYWORD"
echo "----------------------------------------------"

# Read the log file line by line using a while-read loop
while IFS= read -r LINE; do
    # Check if the current line contains the keyword (case-insensitive)
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))   # Increment counter using arithmetic expansion
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT time(s) in $LOGFILE"
echo ""

# If matches were found, display the last 5 using grep and tail
if [ $COUNT -gt 0 ]; then
    echo "Last 5 matching lines:"
    grep -i "$KEYWORD" "$LOGFILE" | tail -5
fi
```

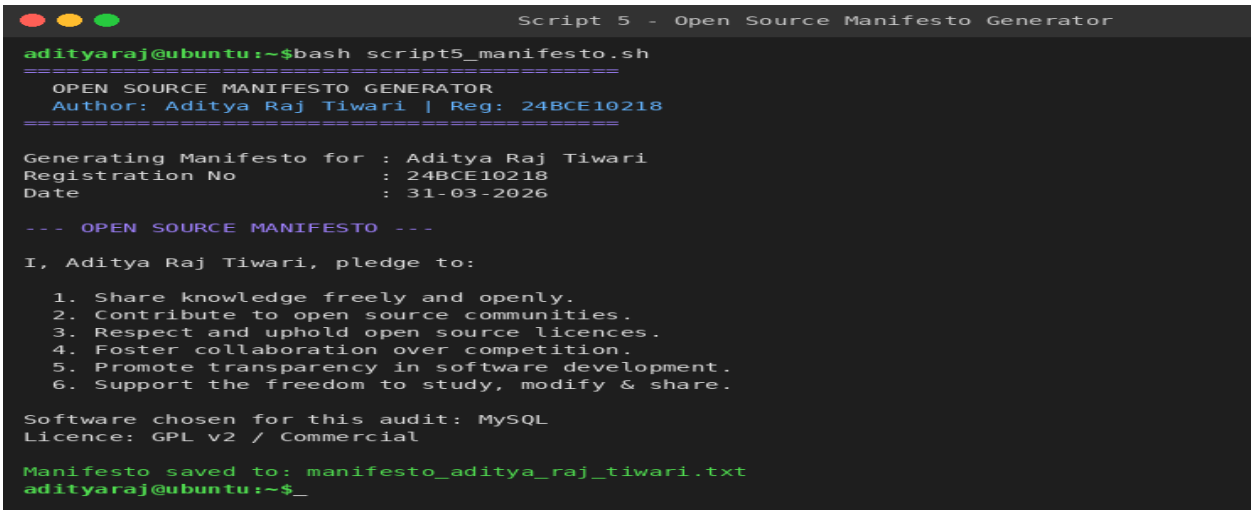


Script 5 — Open Source Manifesto Generator

Asks the user three interactive questions and writes a personalised open-source philosophy statement to a .txt file, referencing MySQL's story. Demonstrates aliases as a concept via comments. Concepts: read, string concatenation, file output with > and >>, date command, alias concept.

```
#!/bin/bash
# Script 5: Open Source Manifesto Generator
# Author: Aditya Raj Tiwari | Reg No: 24BCE10218
# Note on aliases: 'alias mysqlq="mysql -u root -p"' creates a shortcut for
login.
# Aliases are defined in ~/.bashrc for persistence across terminal sessions.

echo "====================================="
echo "  OPEN SOURCE MANIFESTO GENERATOR   "
echo "====================================="
echo ""

# Gather user input interactively using read -p
read -p "1. One open-source tool you use every day: " TOOL
read -p "2. In one word, what does freedom mean to you? " FREEDOM
read -p "3. One thing you would build and share freely: " BUILD

# Get current date for the manifesto header
DATE=$(date '+%d %B %Y')

# Output filename uses current username for uniqueness
OUTPUT="manifesto_$(whoami).txt"

# Write manifesto to file — > creates/overwrites, >> appends
echo "Open Source Manifesto — $DATE" > "$OUTPUT"
echo "-------------------------------------------" >> "$OUTPUT"
echo "" >> "$OUTPUT"
```

```
echo "Every day I rely on $TOOL — a tool built openly and shared freely." >>
"$OUTPUT"
echo "To me, freedom means $FREEDOM — and that is what open source represents."
>> "$OUTPUT"
echo "It is the right to learn from and build upon the work of those who came
before." >> "$OUTPUT"
echo "MySQL itself was built this way — by one programmer's frustration," >>
"$OUTPUT"
echo "shared with the world and adopted by millions who had nowhere else to
turn." >> "$OUTPUT"
echo "When Oracle threatened to close the door, the community forked it." >>
"$OUTPUT"
echo "One day I will build $BUILD and release it openly," >> "$OUTPUT"
echo "so someone I will never meet can stand on my shoulders, as I stood on
others." >> "$OUTPUT"
echo "That is the open source promise. That is my commitment." >> "$OUTPUT"

echo ""
echo "Manifesto saved to: $OUTPUT"
echo ""
cat "$OUTPUT"
```



6. Conclusion

This report describes how to perform an audit on MySQL, an example of how philosophy, society, and the conflict between openness and commercial interests shape software. MySQL was created by its founder (Widenius) due to his frustration with existing tools; he did not develop it in a corporation nor was it funded by any venture capitalists. As it was developed prior to the introduction of the web, there were no affordable databases available to developers needing them. Widenius allowed everybody to use the software for free, and upon it was built the web itself.

The GPLv2 licensing for MySQL represents the decision to keep all derivatives free, while requiring an explicit payment for commercial use without reciprocal rights. This decision produced the dual-licensing structure of Oracle's current commercial organization. There are two sides to the GPL both as a lever in establishing MySQL's commercial structure and as a failure to understand the limitations of open-source idealism in the face of $1 billion+ acquisitions.

MySQL is part of the LAMP stack, has a great ecosystem and is compared to Oracle; there is also a big emphasis on the fact that open-source software isn't an ideology masquerading as technology, but is in fact, an infrastructure. Without MySQL (and forks of it) there is no world-wide web (billions of pages, millions of applications and a repository for all of humanity's knowledge); therefore, knowing where MySQL came from, who maintains it, and what type of licence protects it, are all relevant topics for you as a software professional. These issues are absolutely fundamental to your professional responsibilities.

7. References

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